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Student Number

Sydney Girls High School

2020

TRIAL
HSC
EXAMINATION

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used

Total Marks: 100

Section I – 20 marks (pages 2 – 11)

- Attempt Questions 1-20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 13 - 32)

- Attempt Questions 21 - 30
- Allow about 2 hours and 25 minutes for this section

Section I

20 marks

Attempt Questions 1 - 20

Allow about 35 minutes for this part.

Use the multiple-choice answer sheet for Questions 1–20.

1. First-line defences that mammals have against invasion by disease-causing bacteria include
 - A. lysozymes.
 - B. interferons.
 - C. antibodies.
 - D. killer T cells.

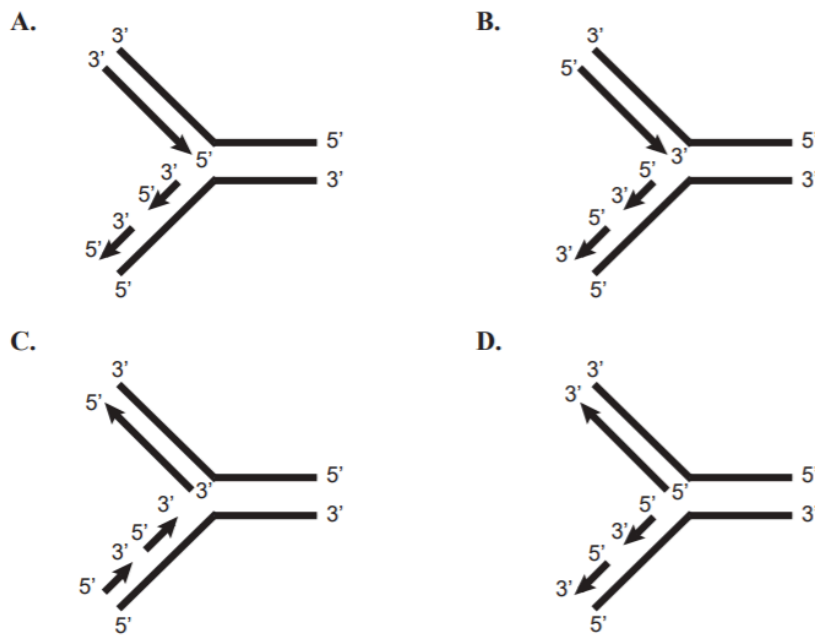
2. Diphtheria is a disease caused by the bacterium *Corynebacterium diphtheriae*.
A six-month old baby boy, whose mother and father both had diphtheria as children, will develop active immunity against diphtheria if he
 - A. is being breast-fed by the mother.
 - B. receives a blood transfusion from the father.
 - C. receives an injection of dead diphtheria bacteria.
 - D. receives injections of gamma globulin from the mother.

3. In sheep, white wool (W) is dominant to black wool (w). In 2005, a farmer with a flock of sheep containing a few black sheep decided to prevent all black sheep from breeding in the future. White sheep in the flock continued to reproduce in that year and the years that followed.

What could reasonably be predicted as a result of the farmer's action?

- A. no black lambs were born in 2006 and beyond.
- B. black lambs were born in 2006 but none in the following years.
- C. the proportion of black lambs born in 2006 was the same as in previous years.
- D. the proportion of black lambs born would decline gradually in the years following 2005.

4. The structure of a DNA replication fork is best represented by which of the following?

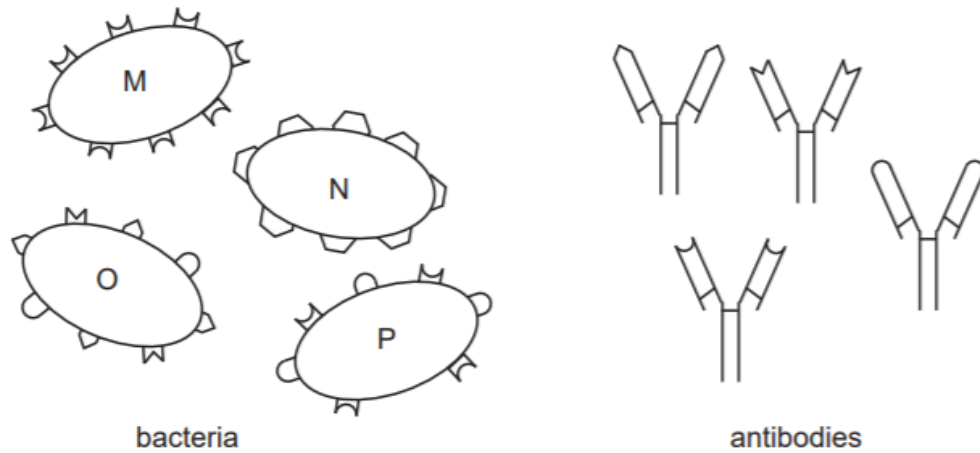


5. *Lucilia cuprina*, the sheep blowfly, lays its eggs in wounds and the wet fleece of sheep. The larvae hatch and burrow into the sheep's skin, causing distress, reduced wool production and sometimes death. Particular chemicals were used in the past to control the *L. cuprina* but these became less effective as sheep blowfly developed a resistance to the chemicals.

The cause of the increased resistance to the chemicals was most likely due to

- A. farmers successively reducing the levels of insecticide applied to sheep.
- B. the insecticide producing a change in a gene which enhanced the survival of the blowfly.
- C. a chance mutation in a blowfly gene conferring a survival advantage in the chemical environment.
- D. the insecticide producing a change in phenotype which enhanced reproduction of the blowfly.

6. A young woman stood on a dirty, rusty nail. The following diagrams show bacteria isolated from the wound and a range of antibodies that were already present in her body.



Which of the microorganism is most likely to cause a severe infection?

- A. M.
- B. N.
- C. O.
- D. P.

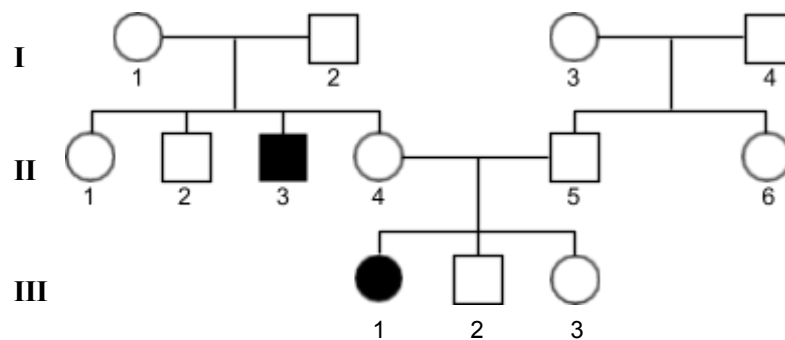
Use the following diagram to answer Questions 7 – 8.

Phenylketonuria (PKU) is a condition in which affected individuals fail to produce the enzyme phenylalanine hydroxylase. PKU is inherited as an autosomal recessive condition and the alleles involved are:

N : normal production of the enzyme

n : lack of enzyme

The following pedigree shows a family in which two members have PKU.



7. In the pedigree shown, individuals that must be heterozygous for PKU include

- A. I-2
- B. I-4
- C. II-1
- D. II-6

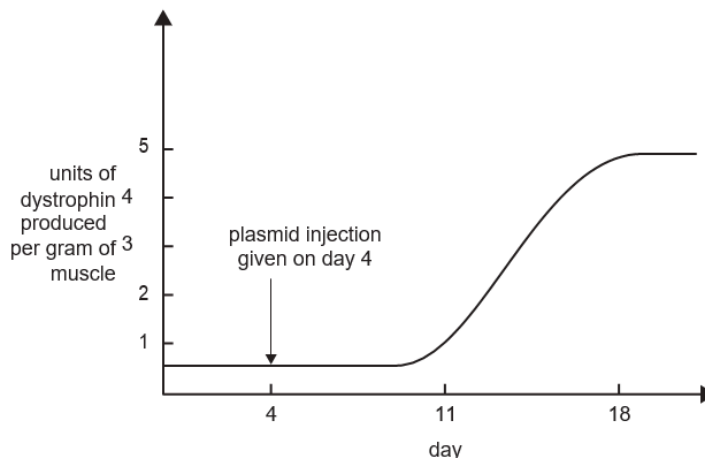
8. Of the unaffected individuals, what is the chance that individual III-3 has genotype Nn with regard to the PKU locus?

- A. 1/3
- B. 1/2
- C. 2/3
- D. 3/4

Use the following diagram to answer Questions 9 – 10.

Some mice suffer from a disease called Duchenne muscular dystrophy, where the muscles waste away due to inadequate production of a protein called dystrophin. Researchers isolated the gene for dystrophin production and then inserted it into a plasmid extracted from a bacterium. A concentrated solution of these recombinant plasmids was then injected into the muscles of diseased mice.

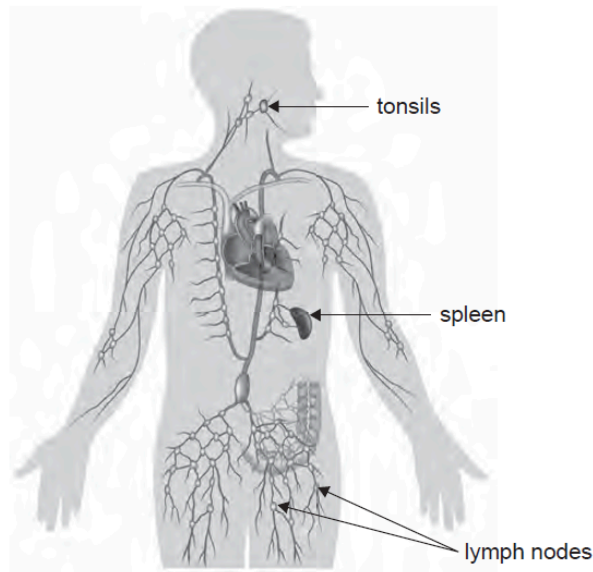
A graph of dystrophin production in these treated mice is shown below.



9. What is the function of the plasmids in this disease treatment?
- A. stimulate a humoral immune response to the bacteria in the treated mice.
 - B. deliver the dystrophin protein directly to the muscles of the treated mice.
 - C. act as a vector for the delivery of dystrophin genes to the muscle cells of the treated mice.
 - D. create a strain of genetically engineered bacteria that will cure Duchenne muscular dystrophy.
10. During the preparation of the recombinant plasmids, researchers used the enzyme DNA ligase. What is the function of DNA ligase in this process?
- A. manufacture an antigen that will be recognised by T-helper cells.
 - B. join the dystrophin gene to the plasmid DNA at complementary sticky ends.
 - C. clone the plasmid in order to produce enough plasmids to ensure effective treatment.
 - D. cut the DNA of the plasmid and the dystrophin gene in the same manner in order to produce matching sticky ends.

11. Below is a diagram of the human lymphatic system.

The human lymphatic system



Source: Alila Medical Media/Shutterstock.com

The lymphatic system includes the lymph nodes, spleen and tonsils. In these particular organs

- A. clotting factors are inactivated to help seal a wound.
- B. clonal selection and proliferation of B cells occurs.
- C. non-self antigens are identified by red blood cells.
- D. the initial response to an allergen is triggered.

12. In the 18th century, farmer Robert Bakewell separated large, fine-boned sheep with long, shiny wool from his native stock to interbreed for future sheep flocks.

This is an example of

- A. genetic fitness.
- B. natural selection.
- C. selective breeding.
- D. allopatric speciation.

13. DNA profiling, using short tandem repeats (STR) within a person's DNA, helps to determine the genetic relationship between individuals.

DNA profiles based on four STRs for five individuals are shown below. The results of a gender identifier are also shown.

STR	Individual 1	Individual 2	Individual 3	Individual 4	Individual 5
CSF1PO	7,14	7,11	8,13	7,14	7,14
TPOX	6,10	10,12	6,9	10,12	10,10
D21S11	27,30	29,32	27,27	29,30	27,28
D8S1179	9,11	12,13	17,17	11,12	9,11

	Individual 1	Individual 2	Individual 3	Individual 4	Individual 5
Gender identifier	male	female	female	male	male

Which one of the following conclusions can be made using the information given?

- A. Individual 4 is the father of Individual 5.
- B. Individual 3 is the mother of Individual 4.
- C. Individual 5 could be the child of Individual 1 and Individual 3.
- D. The parents of Individual 4 could be Individual 1 and Individual 2.

14. Colour blindness is inherited as an X-linked recessive condition. Which of the following statements is a reasonable claim?

- A. colour-blind female must have a colour-blind father.
- B. colour-blind male must have a colour-blind mother.
- C. colour-blind male must have a colour-blind grandfather.
- D. colour-blind female must have a colour-blind grandmother.

15. Genetic drift is

- A. most evident in large populations.
- B. the gene flow between populations.
- C. random changes in the gene pool of a population.
- D. the gradual change in phenotypic frequency resulting from natural selection.

16. Four organisms or agents, and the diseases they cause, are listed in the following table.

causal organism or agent	resultant disease
Prion	Kuru
<i>Staphylococcus</i> bacteria	Boils
<i>Poliomyelitis</i> virus	Poliomyelitis
<i>Trichophyton</i> fungus	Athlete's foot

Which of the diseases can be treated with antibiotics?

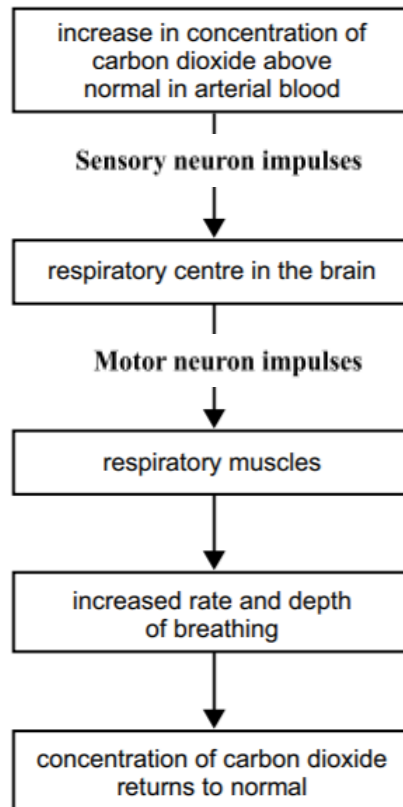
- A. kuru.
- B. boils.
- C. poliomyelitis.
- D. athlete's foot.

17. Which of the following best describes the Major Histocompatibility Complex (MHC) class 1 molecules?

- A. release complement proteins.
- B. are found only on B and T cells.
- C. present foreign antigens to B and T cells.
- D. produce antibodies that are specific to each antigen.

Use the following diagram to answer Questions 18 – 19.

The following diagram shows the regulation of carbon dioxide concentration in arterial blood.



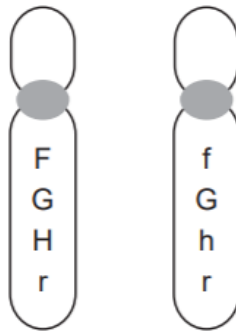
18. Which is the effector in the regulation of carbon dioxide concentration?

- A. respiratory muscles.
- B. motor neuron impulses.
- C. respiratory centre in brain.
- D. sensory neuron impulses.

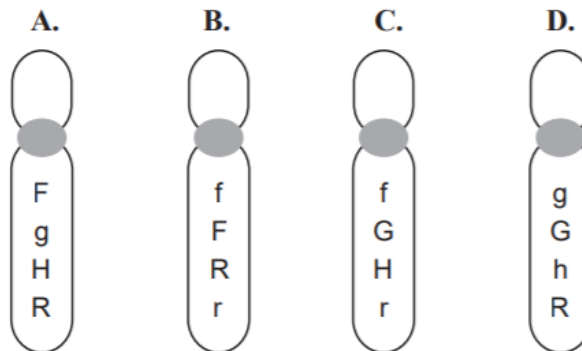
19. What could reasonably be concluded to happen if the concentration of carbon dioxide returns to normal?

- A. respiratory muscles stop contracting.
- B. rate and depth of breathing decrease.
- C. number of afferent nerve impulses increase.
- D. respiratory centre in the brain fails to send sensory neuron impulses.

20. A pair of homologous chromosomes involved in meiosis in an ovary carries the alleles shown.



Assuming no mutations occur during meiosis, chromosomes detected in the eggs produced would include which of the following alleles?



Please turn over

2020

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Student Number

Biology

Section II Answer Booklet

80 marks

Attempt Questions 21- 30

Allow about 2 hours and 25 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of the response.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- If you require more space than is available in the extra writing space provided, please ask a supervising teacher for extra paper. Clearly indicate which question you are answering on the extra paper.

Question 21 (3 marks)

The Arctic fox (shown in the photograph below) lives in the Arctic tundra, which is one of the coldest environments on Earth.

3K



Explain how ONE structural and ONE physiological adaptation enables mammals living in cold environments to maintain a constant core body temperature. Identify clearly in your answer which is the structural feature and which is the physiological process.

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Question 22 (3 marks)

A long-term study of adult and nestling (too young to leave the nest) barn swallows *Hirundo rustica* from Chernobyl were found to have a higher incidence of abnormalities compared with those in uncontaminated areas in Ukraine. Chernobyl was contaminated by nuclear radiation in 1986.

Provide a plausible explanation for the higher incidence of abnormalities observed in the barn swallows that lived in the contaminated area.

3K

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Question 23 (3 marks)

During your study of genetics, you were required to construct a model of polypeptide synthesis.

Discuss the advantages and limitations of your model.

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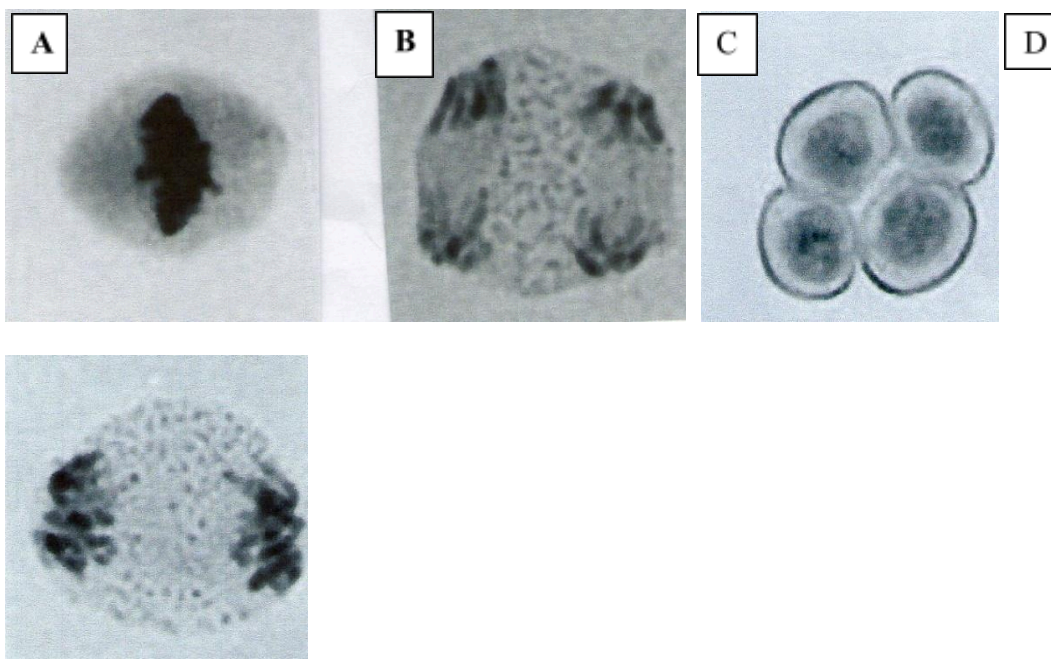
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Question 24 (8 marks)

The following photographs show various steps in the process of meiosis.



- a) Write the correct sequence of letters (A-D) that would show the order in which the steps occur during meiosis.

1A

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- b) Explain ONE way in which variation in gametes can occur during meiosis.

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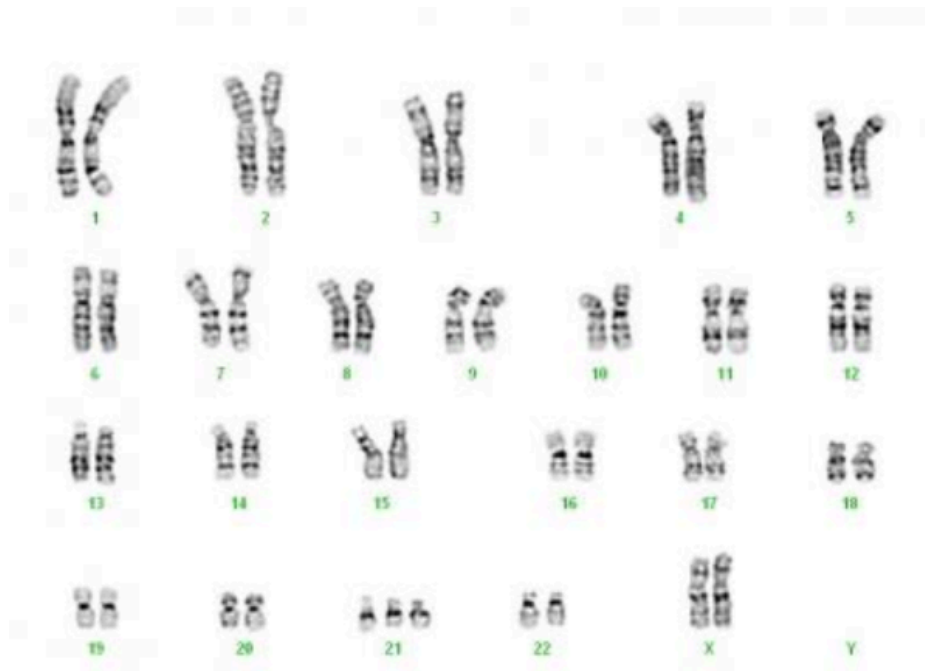
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The karyotype of a human baby is depicted below.



c) Explain how this karyotype could have arisen due to mutation followed by fertilisation.

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Question 25 (14 marks)

Between 2008 and 2012, all Australian states and territories experienced their largest pertussis (whooping cough) epidemic since national reporting began in 1991. Pertussis is a highly contagious respiratory infection caused by the bacteria *Bordetella pertussis*. Pertussis vaccine is offered as part of an immunisation program for children at two months, four months, six months, four years and in Year 10 of secondary school.

- a)** Name the cell that is responsible for the production of antibodies. **1K**

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Two children have been immunised according to the schedule. One is two months old and the other is four months old.

- b)** What difference would there be in the children's levels of antibodies against *Bordetella pertussis*?

1K

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Consider a Year 10 student. Memory cells will have been produced during the periods of immunisation when the student was younger.

- c)** What are TWO advantages of having these memory cells when the student receives their immunisation in Year 10? **2K**

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In NSW, between September 2018 and January 2019, the number of cases of pertussis had increased dramatically. Of the total of 1300 reported cases of pertussis; 55 per cent of cases were adults and most of these adults had been immunised in childhood.

- d)** State one possible reason for the high percentage of adults with pertussis during this period. **1K**

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The human immune response to antigens of *Bordetella pertussis* can be measured by the level of antibodies in the blood.

e) Is this test a measure of innate or adaptive immunity? Explain your answer. **2K**

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f) Explain how vaccination can effectively reduce both the incidence and spread of one named infectious disease through a community. **4K**

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g) Identify and describe ONE other procedure that can be used to prevent the spread of a named infectious disease. **3K**

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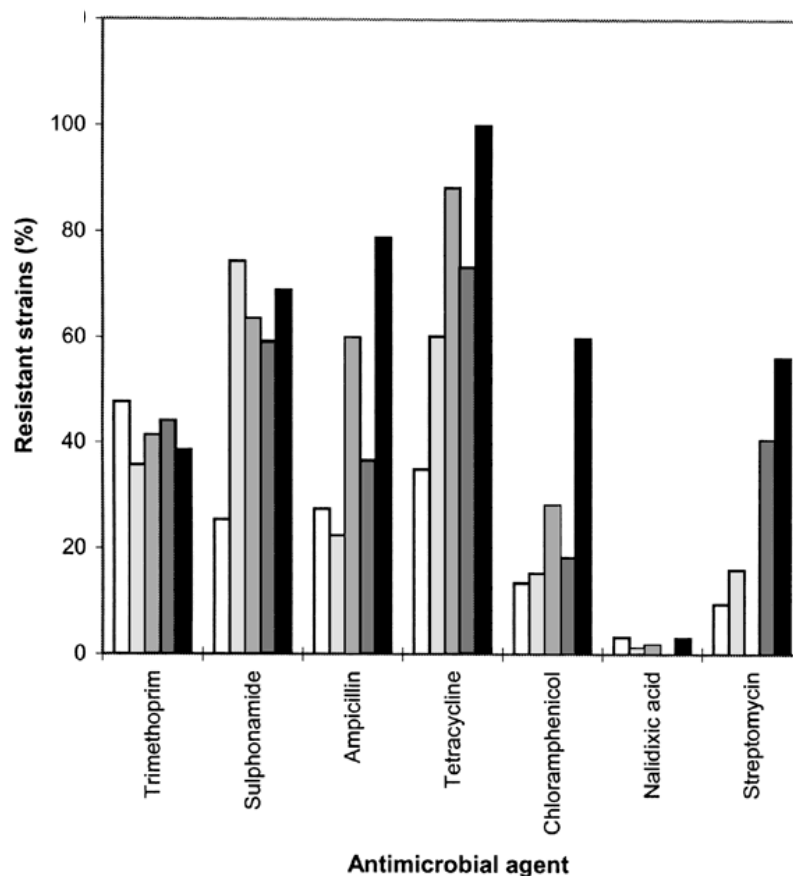
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Question 26 (12 marks)

A student is required to collect data in order to evaluate the effectiveness of antibiotics (also called antimicrobials) since their development in World War II. She finds the following graph in a reputable science journal.

The graph shows percentage of resistant organisms in populations of bacteria at five sampling times after widespread use of named antibiotics (antimicrobials) in the community.



Key:

a) Identify the role of antibiotics in the treatment of infectious diseases.

2K

b) Assess the validity of the secondary data, to the student's investigation. **2K**

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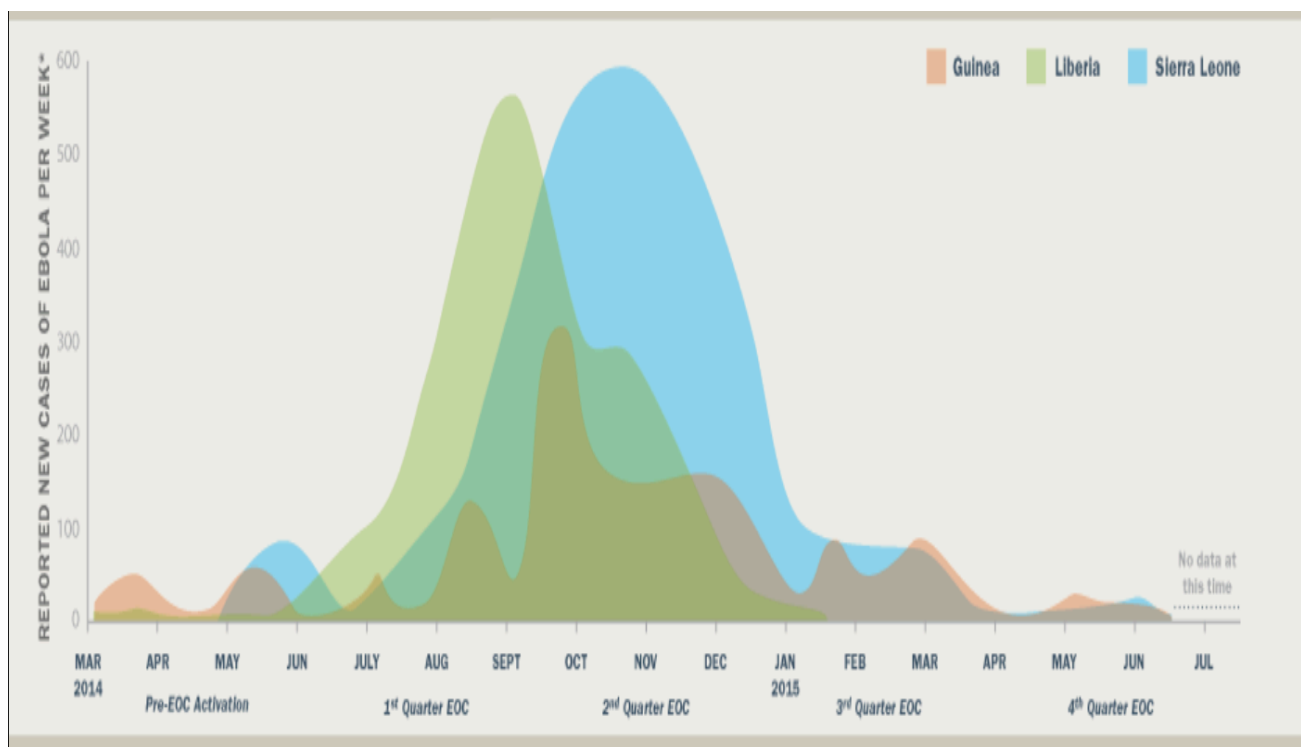
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Question 27 (9 marks)

Ebola, previously known as Ebola hemorrhagic fever, is a rare, extremely contagious and deadly disease caused by infection with one of the Ebola virus strains of the family *Filoviridae*, genus *Ebolavirus*. It is, however, easily preventable. It spreads when people have close contact with body fluids and mucous membranes (as well as items contaminated by these) from infected individuals, including sexual transmission.

The 2014 Ebola epidemic is the largest in history, affecting multiple countries in West Africa (Guinea, Liberia, Sierra Leone). The Centre for Disease Control's (CDC) response to the Ebola epidemic was the largest in the agency's history. Approximately 1,450 CDC responders were deployed (sent to) to Guinea, Liberia, and Sierra Leone from the start of the response in July 2014 to the end of the response at the end of March 2016, including 455 persons with repeat deployments.

The following graph shows the timeline of the repeated control measures put in place by the Centre for Disease Control (CDC).



<https://www.cdc.gov/about/ebola/timeline.html>

EOC- Emergency Operations Center, is a central command and control facility responsible for carrying out the principles of emergency preparedness and emergency management, or disaster management functions.

a) Describe the effect of the control measures put in place in the case of Ebola.

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b) Evaluate environmental management and quarantine methods used to control an epidemic or pandemic using a named infectious disease as an example.

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Question 28 (4 marks)

In diverse habitats, plants display a variety of mechanisms that allow water balance to be maintained.

Explain TWO mechanisms in plants that assist with water balance.

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Question 29 (7 marks)

A number of biotechnologies are used in agriculture. Describe TWO of these technologies and evaluate the effect on biodiversity of using the technologies.

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Question 30 (17 marks)

Albinism is an inherited trait that results in a lack of colour in the eyes and fur of mammals, including guinea pigs. Non-albino guinea pigs have coloured fur.

- a) A non-albino male and a non-albino female guinea pig were crossed and produced a litter containing some albino and some non-albino offspring. Explain how non-albino guinea pigs can produce albino offspring. **3A**

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- b) Both male and female albino offspring were produced in the cross described in part a). On this basis, explain why albinism cannot be a sex-linked trait in guinea pigs. **3A**

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c) Albino guinea pigs are common in captive populations but rare in wild populations. Explain possible reasons for this difference.

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d) Construct a Punnett square to show how a breeder of guinea pigs could ensure that half the offspring were albino. Indicate clearly the genotypes and phenotypes of the offspring.

4A

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- e) A breeder kept only albino guinea pigs. The breeder put one female and two male guinea pigs in the same enclosure. The female had a litter of offspring. The breeder wanted to know which of the male guinea pigs was the father of the litter.

Outline how biotechnology could be used to determine the father of the litter.

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